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R.E.A.R.M. Recognition-Enabled Automated Replacement Manipulator

Embedded Systems II - Inf204

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Agenda

- Problem Statement
- Solution
- Requirements Engineering
- Work Packages
- Milestones
- Roles
- Challenges

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Problem Statement – Wasted Human Potential

The cost of repetition: Highly trained humans in various industries such as logistics and hospitals waste valuable time on repetitive “pick-and-place” sorting tasks (OMRON, 2024)

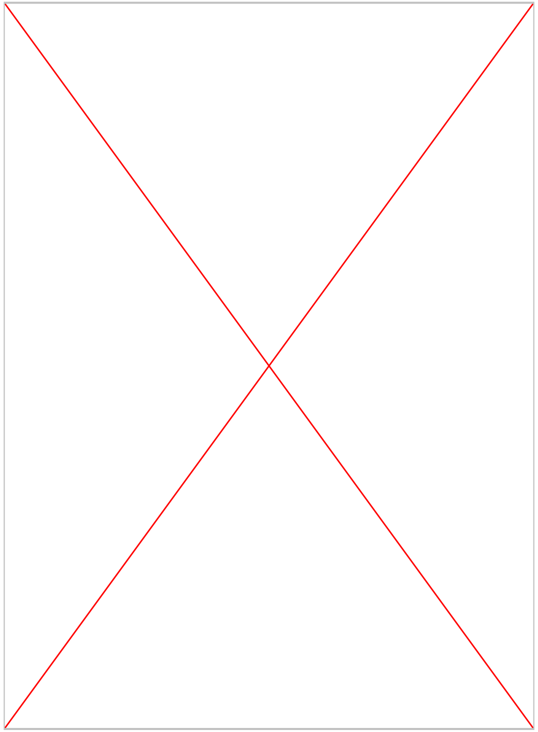
- **General Workforce:** Across various industries, employees spend approximately 62% of their workday on mundane, repetitive tasks rather than high-value work (Asana, 2021).
- **Logistics:** Manual picking is inefficient, impacting 55% of total warehouse operating costs (de Koster et al., 2007)
- **Healthcare:** Healthcare professionals lose nearly 28 hours a week to repetitive, non-clinical tasks (Google Cloud, 2024)

Solution – R.E.A.R.M.

Recognition-**E**nabled **A**utomated **R**eplacement **M**anipulator

- 5 degrees of freedom
- Real-time object detection
- Automatic dimension measurements
- Best-fit grab
- State recovery

https://www.youtube.com/shorts/ifVpf7IHtSo



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Requirement Engineering – Functional

Functional requirements define “what the system must do”.

The R.E.A.R.M must:

1. Detect objects inside the predefined pickup area
2. Determine each object’s position, orientation, and approximate dimensions
3. Calculate and execute a suitable gripping motion
4. Return to a safe home position after each pick operation

Source: Prof. Dr. V. Klös, Embedded Systems I, WiSe 25/26

Requirement Engineering – Non-Functional

These describe how the system operates, such as “When does the system respond?” or “how well the system does it?”

1. Detect and locate objects within 3 seconds after it is placed in the pickup area
2. Achieve at least 90% successful picks for supported objects
3. Not operate outside the predefined pickup area
4. Be capable of operating repeatedly without manual re-calibration

Source: Prof. Dr. V. Klös, Embedded Systems I, WiSe 25/26

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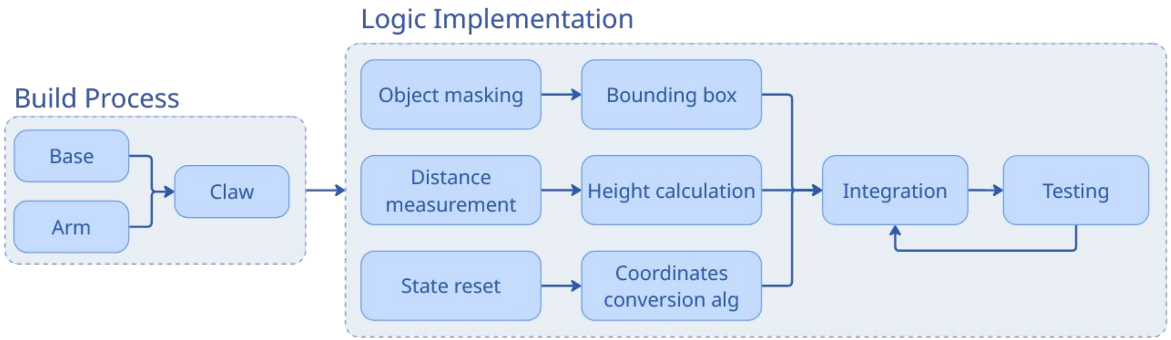
Work Packages

1. Build the robot (base, arm, claw)
2. Implement program logic
   - Object position and dimensions (bounding box, height)
   - Motor logic (state reset, general movement)
   - Integration (intercommunication)
3. Testing & Adjustments
   - Accuracy of the movements
   - Object position & dimensional accuracy

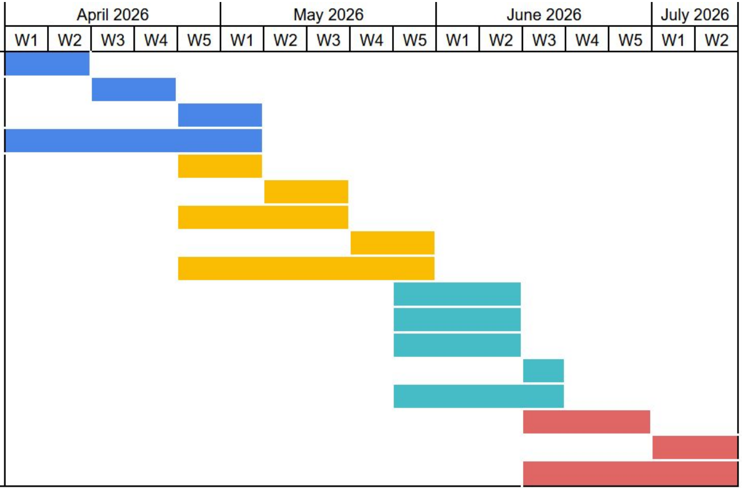


Milestones

1. Planning (05.05.2026)
    - Brainstorming
    - Gantt chart
    - Sketches/3D model
2. Build (26.05.2026)
    - Base
    - Arm
    - Claw
3. Software (16.06.2026)
    - Object detection & dimension measurements
    - Coordinates conversion algorithm
    - Integration
4. Final Product (07.07.2026)

Milestones

No	Task Name	Assigned members
	Brainstorming	all Members of the team
	Sketches / 3D model	Rafael
	Gantt chart	all Members of the team
M1	**Planning (05.05.2026)**	
	Base	Rostik & Amir
	Arm	Rostik & Amir
	Claw	Rostik & Rafael
	Integration	all Members of the team
M2	**Build (26.05.2026)**	
	Object detection	Amir
	Dimension measurements	Rafael
	Coordinates conversion algorithm	Rostik
	Integration	all Members of the team
M3	**Software (16.06.2026)**	
	Adjustments & Testing	all Members of the team
	Preparation for the final presentation	all Members of the team
M4	**Final Product (07.07.2026)**	



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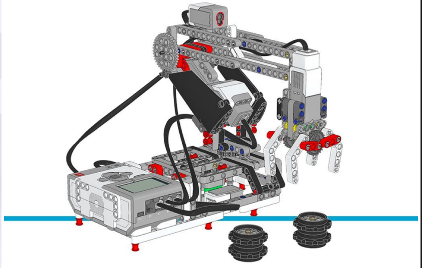
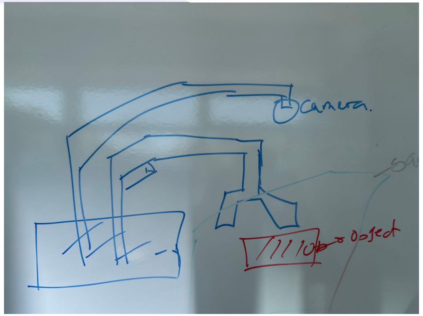
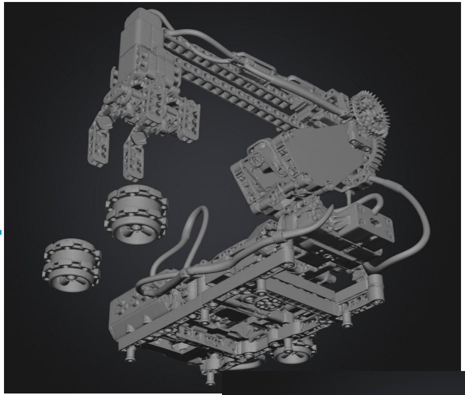
Sketches

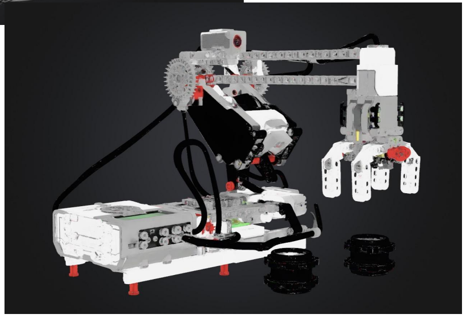
Ideation





3D Model - Link





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Roles & Assignments

1. Amir → Integration Manager
   - Object Detection
2. Rafael → Project manager & Quality Manager
   - Dimension measurements
3. Rostik → Technical Manager
   - Motor Controls

Challenges

- Component Management (partially not enough pieces)
- Claw Complexity (2-axis linear motion in a compact build)
- Coordinates Conversion Algorithm (non-linearity)
- Motor Accuracy

References

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2. Asana. (2021). *Anatomy of work global index*. Asana. https://asana.com/resources/anatomy-of-work
3. de Koster, R., Le-Duc, T., & Roodbergen, K. J. (2007). Design and control of warehouse order picking: A literature review. *European Journal of Operational Research*, *182*(2), 481–501. https://ideas.repec.org/p/ems/eureri/7322.html
4. Google Cloud. (2024, October 22). *New study from Google Cloud and The Harris Poll reports how generative AI can tackle administrative burdens in U.S. healthcare*. PR Newswire. https://www.prnewswire.com/news-releases/new-study-from-google-cloud-and-the-harris-poll-reports-how-generative-ai-can-tackle-administrative-burdens-in-us-healthcare-302278820.html
5. Klös, V. (2025). *Embedded systems I: Introduction* [PowerPoint slides]. Carl von Ossietzky Universität Oldenburg.

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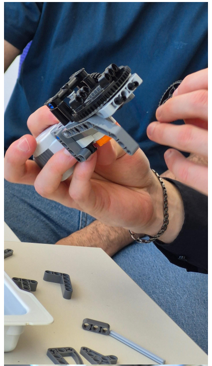
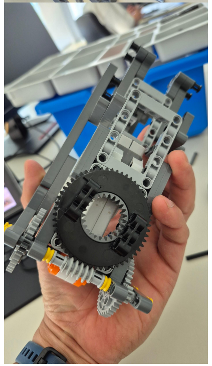
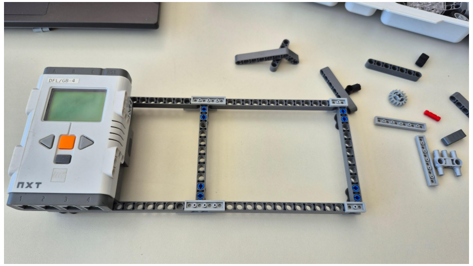
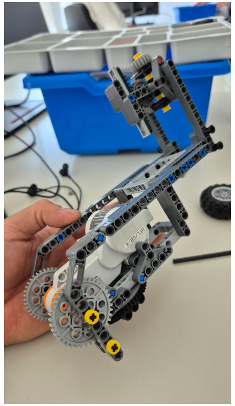
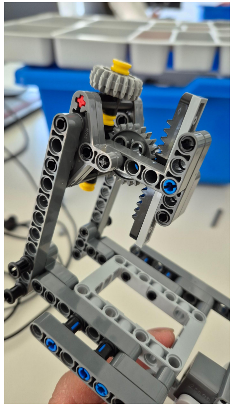
Documentation

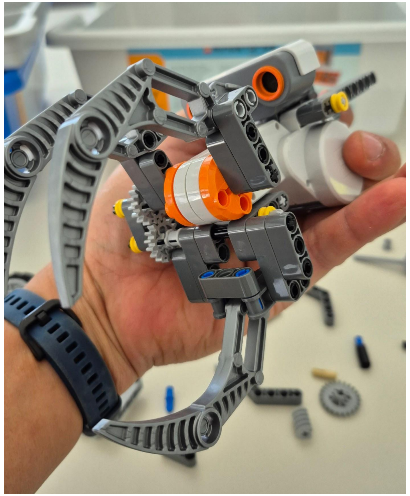


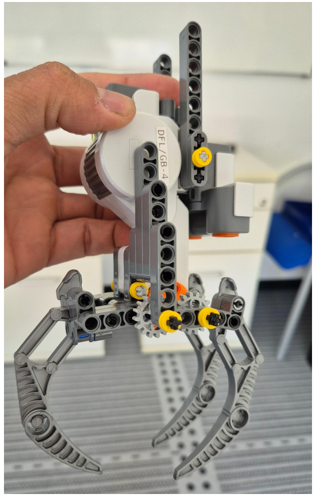


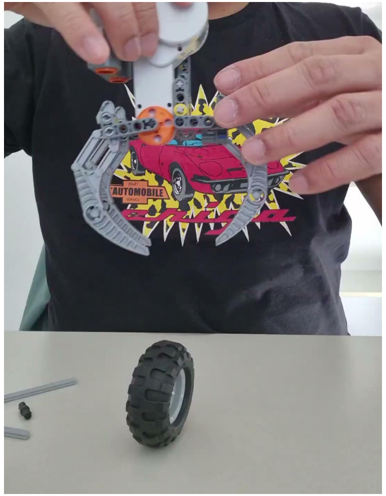












Requirement Engineering – Non-Functional

These describe how the system operates, such as “When does the system respond?” or “How well does the system do it?”

1. **Precision :** High “Accuracy of the movements” and “Motor Accuracy” for physical manipulation
2. **Time Constraints :** “Real time” operation with zero lag between the camera feed and arm movement
3. **Concurrency :** Simultaneous and completely independent execution of visual processing and motor control
4. **Safety and Reliability :** Strict “Internal safety” to prevent malfunctions, dropped object, or collisions

To turn these strict requirements into a reality, we divided the project into specific **work packages.**

Source: Prof. Dr. V. Klös, Embedded Systems I, WiSe 25/26

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Requirement Engineering – Functional

Functional requirements define “what the system must do” . Therefore the R.E.A.R.M must perform:

1. **Visual processing : “**Real time object detection” to locate misplaced items by applying “Object Masking” and generating “Bounding Box”
2. **Spatial processing:** “Automatic dimension measurements” to calculate the exact size and orientation to figure out where the object is by doing “Distance measurement” and “Height calculation”
3. **Manipulation:** a “best-fit grab” by utilizing “5 DoF” to physically move the item
4. **Recover:** “State Recovery” to safely reset the system if an error or drop occurs

Source: Prof. Dr. V. Klös, Embedded Systems I, WiSe 25/26